



SHREEHARI ANBAZHAGAN

DA25C020 | INDIAN INSTITUTE OF TECHNOLOGY MADRAS | DSAI

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EDUCATION

Degree / Examination	Institution / Board	CGPA / %	Year
Joint Masters, Data Science & Artificial Intelligence	IIT Madras & UOB	8.00	2025–27
B.Tech Computer Science & Engineering	SASTRA Deemed University	7.8103	2019–23
Class XII (CBSE)	Maharishi Vidya Mandir Sr. Sec. School	84.8%	2019
Class X (CBSE)	Maharishi Vidya Mandir Sr. Sec. School	9.6	2017

- Distinction-grade marks to date across the Joint Masters, spanning machine learning, data analytics and statistics.

PROFESSIONAL EXPERIENCE

Development Engineer 1 (Data Engineering), Comcast

Jul 2023 – Jun 2025

- Engineered enterprise ETL/ELT pipelines on Kafka, Redshift, PySpark and AWS Glue, cutting job failures and runtime by **60%**.
- Devised automated reconciliation frameworks validating source-to-warehouse data integrity, raising reporting accuracy by **40%**.
- Optimised warehouse consolidation and Redshift snapshot/storage strategy, cutting cloud infrastructure cost by **50%**.
- Deployed anomaly detection with AWS Lookout for Metrics, CloudWatch dashboards and real-time Slack data-quality alerts.
- Migrated legacy reporting jobs to PySpark pipelines and ingested Workday and LDAP sources into the warehouse.
- Led knowledge-transfer sessions and authored the onboarding documentation and runbooks adopted across the team.
- Built EnvSyncer, an internal tool synchronising data across environments to improve downstream availability.

Data Engineering Intern, Comcast

Jan 2023 – Jun 2023

- Built a Slack monitoring and alerting system for ETL runs, with job-status alerts and automated daily cost reports.
- Contributed to the enterprise data warehouse, consolidating multiple upstream sources into a single source of truth.

Traceability Platform Developer Intern, Kahawa Roots (UK) Ltd

May – Jul 2026

- Delivered a farm-to-UK coffee traceability platform end to end as technical lead, mentoring a fellow intern through to launch.
- Implemented AWS Cognito authentication, RBAC, audit logging and secrets management across every platform service.
- Owned regression and deployment testing, CI/CD pipelines, production monitoring and incident debugging to release.

Machine Learning Research Intern, MInt Labs (IIT Madras)

Nov 2025 – Jan 2026

- Modelled real-time bus-ETA for the IITM campus shuttle, engineering **53** spatial-temporal features from GPS streams.
- Built validation, monitoring and fallback handling to keep predictions robust to noisy, real-world GPS data.

PROJECTS

- Open Evidence Map — Master's Thesis** with **KCM** (Kusnierz Consolidated Manufacturing): traces document claims to weighted evidence in a **knowledge graph**, flagging unsupported claims.
Stack: Python, Neo4j, local LLMs (Qwen 2.5 via Ollama), GLiNER NER, FastAPI; plain-English-to-Cypher graph querying.
Features: local-first extraction, human-in-the-loop review, trust × relevance weighting, graph health scoring.
- Kahawa Roots Traceability Platform: offline-first**, farm-to-UK coffee traceability with consumer QR provenance journeys.
Stack: Flutter, FastAPI, Next.js, PostgreSQL on AWS (ECS, RDS, S3, Cognito) via Terraform and GitHub Actions.
Features: offline sync across 9 stations, crash-resilient batch ingest, signed-S3 document uploads, public QR journeys.
- DeltaEd: RAG-powered** AI coach with a growth-based leaderboard and personalised feedback on Google Cloud Run.
Stack: Retrieval-augmented generation with a vector store, LLM APIs, Python, Google Cloud Run (Google hackathon build).
Features: growth-based ranking, personalised RAG coaching loop, cloud-native serverless deployment.
- Bus-ETA Prediction (MInt Labs):** real-time ETA on **17.2M** GPS records at **1.88-minute** MAE with sub-ms inference.
Stack: XGBoost + LightGBM + graph-neural-network ensemble, 53 engineered features, served via FastAPI and Docker.
Features: sub-millisecond inference, model validation, monitoring and fallback handling, robust to noisy real-world GPS data.

COURSEWORK

- Mathematics for Data Science
- Data Structures for Data Science
- Foundation of Machine Learning
- Data Analytics Laboratory
- Current Topics in AI & ML
- Intelligent Interactive Systems
- Natural Language Processing
- Neural Computing
- Image Processing

TECHNICAL SKILLS

- Languages:** Python, SQL, HCL (HashiCorp Configuration Language).
- Machine Learning:** scikit-learn, XGBoost, LightGBM, Graph Neural Networks, RAG, NLP.
- Data Engineering:** ETL/ELT, PySpark, Kafka, Redshift, PostgreSQL, data warehousing, data modelling.
- Cloud & DevOps:** AWS, GCP, Terraform, Docker, CI/CD, Git, GitHub, Linux.
- Tools & Libraries:** Pandas, NumPy, boto3, REST APIs, Pytest, Jupyter, LaTeX.
- Practices:** microservices, event-driven design, data modelling, Agile, code review, documentation.

AWARDS

- Won **Best Pitch** and **Best Use of Google Tools** at a Google Hackathon for DeltaEd, a RAG-powered learning coach.
- Won the **Champ Award** (×2) for outstanding performance and data-engineering delivery at Comcast (Q4 2023, Q2 2024).